Total No. of Questions : 4]

SEAT No. :

PC394

[Total No. of Pages : 3

[6359]-514

S.E. (Electronics/ E & TC) (Insem)

ELECTRICAL CIRCUITS

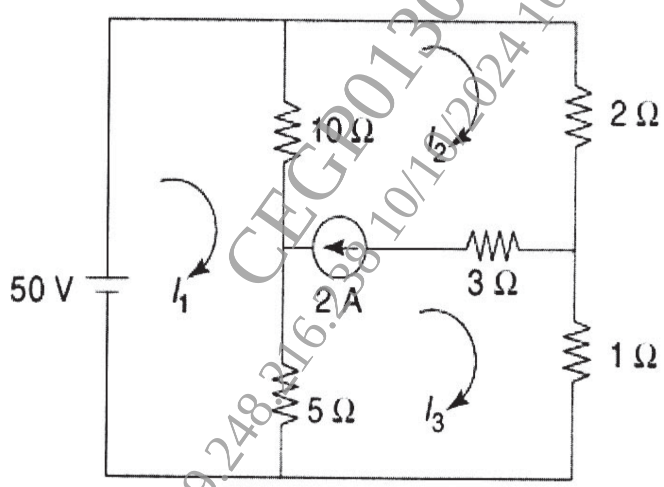
(2019 Pattern) (Semester - III) (204183)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4.***
2) ***Neat diagram and waveforms must be drawn whenever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Use of nonprogrammable calculator is allowed.***
5) ***Assume suitable data, if necessary.***

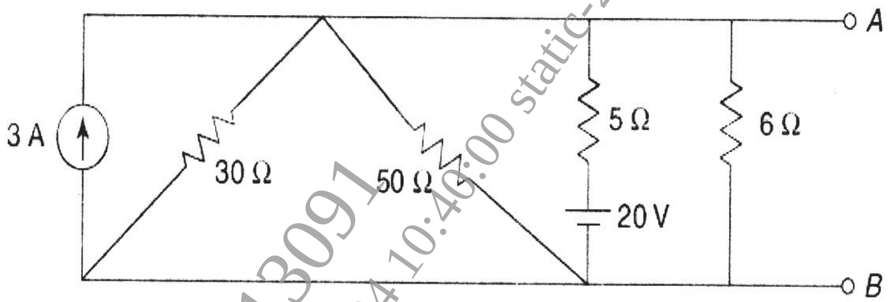
Q1) a) Using super mesh analysis technique determine the current in the 5Ω resistor of the network shown in Fig. **[6]**



b) What is source transformation, explain with neat diagrams. **[3]**

c) Using source transformation technique, replace the circuit between A and B in figure with a voltage source in series with a single resistor. **[6]**

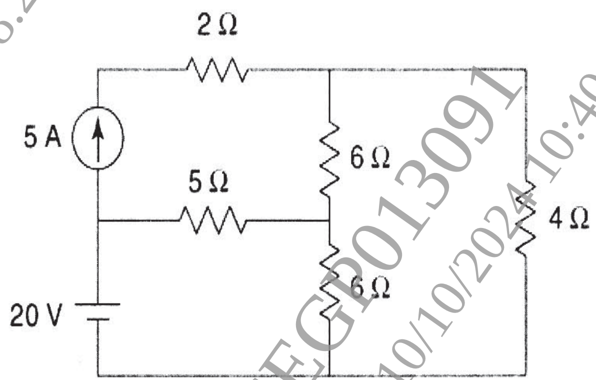
P.T.O.



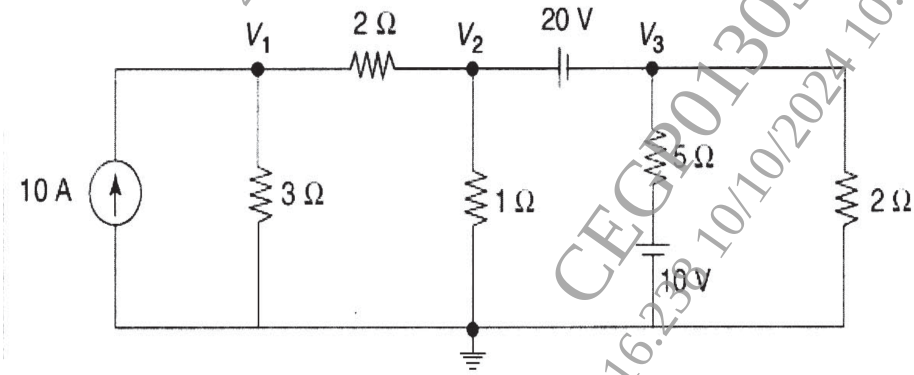
OR

Q2) a) State and explain super position theorem. **[3]**

b) Using super position theorem find the current through the 4Ω resistor. **[6]**

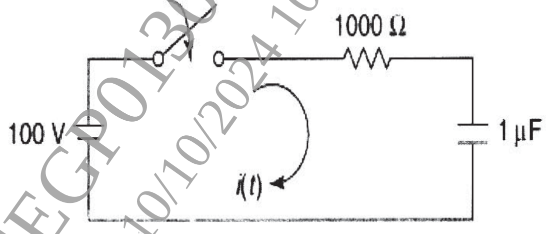


c) What is super node. Determine the current in the 5Ω resistor using super node technique. for the network shown in Fig. **[6]**



Q3) a) Derive the equation for the Complementary Solution for voltage across the capacitor *vc(t)* for Source Free RC Circuit. Draw natural Response of the Circuit for various values of t. **[8]**

b) In the network shown below, the switch is closed at t = 0. With the capacitor uncharged, find value for i, di/dt and d^2i/dt^2 at $t = 0^+$. **[7]**



OR

Q4) a) Obtain the characteristics equation for series R L C Circuit and explain following cases of response with neat diagrams. **[8]**

i) Over damped response

ii) Critically damped response

iii) Under damped response

b) A series R-L circuit is driven by DC Voltage source V. A voltage source is connected in series circuit through a switch K which is closed at t = 0. find expression for current through inductor for t > 0. Sketch the variation of i(t) against time. **[7]**

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